

Relatório Lab.13

Introdução

No laboratório 13 foi possível evidenciar, na prática, como um método de segurança utilizando login, senha e tokens realmente é realizado. Foi implementado um sistema de autenticação usando JWT (JSON Web Token) com quatro módulos: Cliente, Autenticação, Dados protegidos e Cliente_Expirado (para que fosse possível testar o limite do token para 10 segundos antes de sua expiração). O JWT (JSON Web Token) é um padrão aberto (RFC 7519) que define um método compacto e autônomo para transmitir informações de forma segura entre partes como um objeto JSON. Os códigos do laboratório estão em anexo nesse mesmo zip (para encurtar o relatório).

Desenvolvimento

Uma série de prints de terminais e prints do wireshark foram tirados para uma análise mais minuciosa, a partir de agora as explicações serão feitas a partir do cabeçalho dos prints, separados por tópicos (de acordo com o roteiro).

Login Bem-sucedido

A partir dos módulos criados citados na introdução, foi feito diversas requisições a partir de um cliente, que tinha o objetivo de realizar inputs para autenticações e verificar se o processo foi bem-sucedido ou não (logo mais veremos que foi bem-sucedido).

PRINT 1: É possível verificar no print abaixo que o teste 1 (tentativa de login) foi bem-sucedido (status 200 OK). Inclusive com a resposta sendo o token que foi gerado pela ferramenta JWT.

```
source /home/allan-samuel/python/venv/bin/activate
allan-samuel@allan-samuel-VirtualBox:~/python$ source /home/allan-samuel/python/venv/bin/activate
(venv) allan-samuel@allan-samuel-VirtualBox:~/python$ python3 client.py

--- Teste 1: Tentativa de Login para usuariol ---
Status Login: 200
Resposta: {'token': 'eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJleHAiOjE3NjI0Mjg4MDEsImhhdCI6MTc2MjQyODUwMScwIj3ViIjoidXN1YXJpbzEiL0T8TI'}
Token Recebido (Verificado?): Sim

--- Teste 2: Acesso a Dados Protegidos ---
Status Acesso: 200
Resposta: {'data': 'Detalhes de conta e histórico de pedidos do usuariol.', 'msg': 'Acesso concedido para: usuariol'}

--- Teste 4.1: Acesso Sem Token ---
Status Acesso: 401
Resposta: {'msg': 'Token não fornecido'}

--- Teste 4.2: Acesso com Token Inválido (Modificado) ---

--- Teste 2: Acesso a Dados Protegidos ---
Status Acesso: 401
Resposta: {'msg': 'Assinatura do token inválida. Token adulterado ou chave incorreta.'}
(venv) allan-samuel@allan-samuel-VirtualBox:~/python$ python3 client.py
```

PRINT 2: No segundo print (abaixo) é possível verificar que a autenticação foi bem-sucedida (status 200 OK). Feito a partir do código em python da autenticação.

```
source /home/allan-samuel/python/venv/bin/activate
allan-samuel@allan-samuel-VirtualBox:~/python$ source /home/allan-samuel/python/venv/bin/activate
(venv) allan-samuel@allan-samuel-VirtualBox:~/python$ python3 auth_api.py
* Serving Flask app 'auth_api'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 702-603-275
/home/allan-samuel/python/auth_api.py:27: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for r
sent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    tempo_expiracao = datetime.datetime.utcnow() + datetime.timedelta(seconds=app.config['JWT_EXPIRATION_SECONDS'])
/home/allan-samuel/python/auth_api.py:31: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for r
sent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    'iat': datetime.datetime.utcnow(),
127.0.0.1 - - [06/Nov/2025 08:28:21] "POST /login HTTP/1.1" 200 -
(venv) allan-samuel@allan-samuel-VirtualBox:~/python$
```

PRINTS DO WIRESHARK: Os prints do wireshark abaixo demonstram o sucesso do login (pela perspectiva da captura de pacotes em loopback).

/tmp/wireshark_loTTX9E3.pcapng 466 total packets, 212 shown

No.	Time	Source	Destination	Protocol	Length	Info
37	69.478066550	127.0.0.1	127.0.0.1	TCP	74	59838 → 5000 [SYN]
Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=3387980191 TSecr=0 WS=128						
Frame 37: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 59838, Dst Port: 5000, Seq: 0, Len: 0						
38	69.478095850	127.0.0.1	127.0.0.1	TCP	74	5000 → 59838 [SYN, ACK]
Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=3387980191 TSecr=3387980191 WS=128						
Frame 38: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5000, Dst Port: 59838, Seq: 0, Ack: 1, Len: 0						
39	69.478119648	127.0.0.1	127.0.0.1	TCP	66	59838 → 5000 [ACK]
Seq=1 Ack=1 Win=65536 Len=0 TSval=3387980191 TSecr=3387980191						
Frame 39: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 59838, Dst Port: 5000, Seq: 1, Ack: 1, Len: 0						
40	69.478227717	127.0.0.1	127.0.0.1	TCP	269	59838 → 5000 [PSH, ACK]
Seq=1 Ack=1 Win=65536 Len=203 TSval=3387980191 TSecr=3387980191 [TCP segment of a reassembled PDU]						
Frame 40: 269 bytes on wire (2152 bits), 269 bytes captured (2152 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 59838, Dst Port: 5000, Seq: 1, Ack: 1, Len: 203						
41	69.478234603	127.0.0.1	127.0.0.1	TCP	66	5000 → 59838 [ACK]
Seq=1 Ack=204 Win=65280 Len=0 TSval=3387980191 TSecr=3387980191						
Frame 41: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5000, Dst Port: 59838, Seq: 1, Ack: 204, Len: 0						
42	69.478248467	127.0.0.1	127.0.0.1	HTTP/JSON	114	POST /login HTTP/1.1 ,
JSON (application/json)						
Frame 42: 114 bytes on wire (912 bits), 114 bytes captured (912 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 59838, Dst Port: 5000, Seq: 204, Ack: 1, Len: 48						
[2 Reassembled TCP Segments (251 bytes): #40(203), #42(48)]						
Hypertext Transfer Protocol						
POST /login HTTP/1.1\r\n						
Host: 127.0.0.1:5000\r\n						
User-Agent: python-requests/2.32.5\r\n						
Accept-Encoding: gzip, deflate\r\n						
Accept: */*\r\n						
Connection: keep-alive\r\n						
Content-Length: 48\r\n						
Content-Type: application/json\r\n						
\r\n						
[Full request URI: http://127.0.0.1:5000/login]						
[HTTP request 1/1]						
[Response in frame: 46]						
File Data: 48 bytes						
JavaScript Object Notation: application/json						
43	69.478253128	127.0.0.1	127.0.0.1	TCP	66	5000 → 59838 [ACK]
Seq=1 Ack=252 Win=65280 Len=0 TSval=3387980191 TSecr=3387980191						
Frame 43: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5000, Dst Port: 59838, Seq: 1, Ack: 252, Len: 0						
44	69.480882779	127.0.0.1	127.0.0.1	TCP	232	5000 → 59838 [PSH, ACK]
Seq=1 Ack=252 Win=65536 Len=166 TSval=3387980194 TSecr=3387980191 [TCP segment of a reassembled PDU]						
Frame 44: 232 bytes on wire (1856 bits), 232 bytes captured (1856 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5000, Dst Port: 59838, Seq: 1, Ack: 252, Len: 166						
45	69.480892683	127.0.0.1	127.0.0.1	TCP	66	59838 → 5000 [ACK]
Seq=252 Ack=167 Win=65408 Len=0 TSval=3387980194 TSecr=3387980194						
Frame 45: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						

/tmp/wireshark_loTTX9E3.pcapng 466 total packets, 212 shown

```
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 59838, Dst Port: 5000, Seq: 252, Ack: 167, Len: 0
No.      Time                Source                Destination            Protocol Length Info
 46 69.480908862    127.0.0.1            127.0.0.1              HTTP/JSON 253      HTTP/1.1 200 OK , JSON
(application/json)
Frame 46: 253 bytes on wire (2024 bits), 253 bytes captured (2024 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 5000, Dst Port: 59838, Seq: 167, Ack: 252, Len: 187
[2 Reassembled TCP Segments (353 bytes): #44(166), #46(187)]
Hypertext Transfer Protocol
  HTTP/1.1 200 OK\r\n
  Server: Werkzeug/3.1.3 Python/3.12.3\r\n
  Date: Thu, 06 Nov 2025 11:28:21 GMT\r\n
  Content-Type: application/json\r\n
  Content-Length: 187\r\n
  Connection: close\r\n
  \r\n
[HTTP response 1/1]
[Time since request: 0.002660395 seconds]
[Request in frame: 42]
[Request URI: http://127.0.0.1:5000/login]
File Data: 187 bytes
JavaScript Object Notation: application/json
No.      Time                Source                Destination            Protocol Length Info
 47 69.480913583    127.0.0.1            127.0.0.1              TCP          66      59838 → 5000 [ACK]
Seq=252 Ack=354 Win=65280 Len=0 TSval=3387980194 TSecr=3387980194
Frame 47: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 59838, Dst Port: 5000, Seq: 252, Ack: 354, Len: 0
No.      Time                Source                Destination            Protocol Length Info
 48 69.481580368    127.0.0.1            127.0.0.1              TCP          66      59838 → 5000 [FIN, ACK]
Seq=252 Ack=354 Win=65536 Len=0 TSval=3387980195 TSecr=3387980194
Frame 48: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 59838, Dst Port: 5000, Seq: 252, Ack: 354, Len: 0
No.      Time                Source                Destination            Protocol Length Info
 49 69.481939224    127.0.0.1            127.0.0.1              TCP          66      5000 → 59838 [FIN, ACK]
Seq=354 Ack=253 Win=65536 Len=0 TSval=3387980195 TSecr=3387980195
Frame 49: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 5000, Dst Port: 59838, Seq: 354, Ack: 253, Len: 0
No.      Time                Source                Destination            Protocol Length Info
 50 69.481960586    127.0.0.1            127.0.0.1              TCP          66      59838 → 5000 [ACK]
Seq=253 Ack=355 Win=65536 Len=0 TSval=3387980195 TSecr=3387980195
Frame 50: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 59838, Dst Port: 5000, Seq: 253, Ack: 355, Len: 0
```

Acesso a dados protegidos

O segundo objetivo é acessar dados protegidos pelo JWT, foi criado o código em python do dos dados protegidos, chamado data_api.py (está em anexo), logo mais veremos que a tentativa de acessar esses dados protegidos foi um sucesso (mesmo nos códigos de erro, que são específicos do roteiro).

PRINT 1: O print abaixo demonstra as tentativas de obter dados protegidos a partir de determinadas circunstâncias que serão mostradas a seguir. Os códigos de erro são correspondentes às circunstâncias adotadas, sendo 200 OK, 401 para erros de token (que veremos).

```
source /home/allan-samuel/python/venv/bin/activate
allan-samuel@allan-samuel-VirtualBox:~/python$ source /home/allan-samuel/python/venv/bin/activate
(venv) allan-samuel@allan-samuel-VirtualBox:~/python$ python3 data_api.py
* Serving Flask app 'data_api'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5001
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 702-603-275
127.0.0.1 - - [06/Nov/2025 08:28:21] "GET /protected HTTP/1.1" 200 -
127.0.0.1 - - [06/Nov/2025 08:28:21] "GET /protected HTTP/1.1" 401 -
127.0.0.1 - - [06/Nov/2025 08:28:21] "GET /protected HTTP/1.1" 401 -
* Detected change in '/home/allan-samuel/python/auth_api.py', reloading
* Restarting with stat
* Debugger is active!
* Debugger PIN: 702-603-275
127.0.0.1 - - [06/Nov/2025 08:43:34] "GET /protected HTTP/1.1" 200 -
127.0.0.1 - - [06/Nov/2025 08:43:34] "GET /protected HTTP/1.1" 401 -
127.0.0.1 - - [06/Nov/2025 08:43:34] "GET /protected HTTP/1.1" 401 -
127.0.0.1 - - [06/Nov/2025 08:45:53] "GET /protected HTTP/1.1" 200 -
127.0.0.1 - - [06/Nov/2025 08:45:53] "GET /protected HTTP/1.1" 401 -
127.0.0.1 - - [06/Nov/2025 08:45:53] "GET /protected HTTP/1.1" 401 -
* Detected change in '/home/allan-samuel/python/clientexp.py', reloading
* Restarting with stat
* Debugger is active!
* Debugger PIN: 702-603-275
127.0.0.1 - - [06/Nov/2025 08:53:36] "GET /protected HTTP/1.1" 200 -
127.0.0.1 - - [06/Nov/2025 08:53:47] "GET /protected HTTP/1.1" 401 -
127.0.0.1 - - [06/Nov/2025 08:53:47] "GET /protected HTTP/1.1" 401 -
127.0.0.1 - - [06/Nov/2025 08:53:47] "GET /protected HTTP/1.1" 401 -
127.0.0.1 - - [06/Nov/2025 08:53:47] "GET /protected HTTP/1.1" 401 -
127.0.0.1 - - [06/Nov/2025 09:04:12] "GET /protected HTTP/1.1" 200 -
127.0.0.1 - - [06/Nov/2025 09:04:23] "GET /protected HTTP/1.1" 401 -
127.0.0.1 - - [06/Nov/2025 09:04:23] "GET /protected HTTP/1.1" 401 -
127.0.0.1 - - [06/Nov/2025 09:04:23] "GET /protected HTTP/1.1" 401 -
```

PRINT 2: A primeira circunstância é o acesso bem-sucedido, ou seja, o usuário conseguiu pegar as suas informações de uma maneira simples através do login.

```
--- TESTE 2: ACESSO COM TOKEN VÁLIDO: Acesso a Dados Protegidos ---
Tentando acesso COM TOKEN...
Status Acesso: 200
Resposta: {'data': 'Detalhes de conta e histórico de pedidos do usuario1.', 'msg': 'Acesso concedido para: usuario1'}
```

PRINT 3: A segunda circunstância é o token não fornecido, ou seja, a tentativa de acessar os dados protegidos sem um token:

```
--- Teste 4.1: Acesso Sem Token ---
Status Acesso: 401
Resposta: {'msg': 'Token não fornecido'}
```

PRINT 4: A terceira circunstância é o token inválido, tentar acessar os dados protegidos com um token inválido.

```
--- Teste 4.2: Acesso com Token Inválido (Modificado) ---  
  
--- Teste 2: Acesso a Dados Protegidos ---  
Status Acesso: 401  
Resposta: {'msg': 'Assinatura do token inválida. Token adulterado ou chave incorreta.'}  
• (venv) allan-samuel@allan-samuel-VirtualBox:~/python$ python3 client.py
```

PRINT 5: O print abaixo mostra os testes feitos pelo cliente a partir de um token que não expira com facilidade (veremos um token que expira em 10 segundos no tópico Circunstância Peculiar).

```
source /home/allan-samuel/python/venv/bin/activate
• allan-samuel@allan-samuel-VirtualBox:~/python$ source /home/allan-samuel/python/venv/bin/activate
• (venv) allan-samuel@allan-samuel-VirtualBox:~/python$ python3 client.py

--- Teste 1: Tentativa de Login para usuario1 ---
Status Login: 200
Resposta: {'token': 'eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJleHAiOjE3NjI0Mjg4MDEsImhhdCI6MTkzMjQyODUuMScwV3ViIjoiaW9kXWlYXjpbzEiLCJ08TII'}
Token Recebido (Verificado?): Sim

--- Teste 2: Acesso a Dados Protegidos ---
Status Acesso: 200
Resposta: {'data': 'Detalhes de conta e histórico de pedidos do usuario1.', 'msg': 'Acesso concedido para: usuario1'}

--- Teste 4.1: Acesso Sem Token ---
Status Acesso: 401
Resposta: {'msg': 'Token não fornecido'}

--- Teste 4.2: Acesso com Token Inválido (Modificado) ---

--- Teste 2: Acesso a Dados Protegidos ---
Status Acesso: 401
Resposta: {'msg': 'Assinatura do token inválida. Token adulterado ou chave incorreta.'}
• (venv) allan-samuel@allan-samuel-VirtualBox:~/python$ python3 client.py
```

Circunstância Peculiar

Temos uma circunstância peculiar: e se o cliente tentar acessar seus dados protegidos em uma situação em que o token expira em 10 segundos? Ou seja, o cliente tentará recuperar seus dados depois de esperar 11 segundos, então não conseguirá porque o token de proteção expirou.

PRINT 1: O print abaixo mostra a tentativa do cliente de acessar os seus dados protegidos depois que seu token (que garante a segurança de seus dados) expirou.

```
=====
--- TESTE 3: TOKEN EXPIRADO (Espera de 11s) ---
=====
1. Aguardando 11 segundos para o token expirar...

--- Teste 3: Acesso Pós-Expiração: Acesso a Dados Protegidos ---
Tentando acesso COM TOKEN...
Status Acesso: 401
Resposta: {'msg': 'Token expirado. Por favor, faça login novamente.'}
```

PRINT 2: O print abaixo mostra os testes completos do cliente a partir de um token que expira em 10 segundos (TESTE 3).

```
=====
--- TESTE 1: LOGIN BEM-SUCEDIDO (usuariol) ---
=====
Status Login: 200
Resposta: {'token': 'eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJleHAiOjE3NjI0MzA2NjIsImhhdCI6MTc2MjQzMDY1Miwic3ViIjoidXNlYXJpbzEiLCJ3_rqI'}
Token Recebido (Verificado?): SIM

--- TESTE 2: ACESSO COM TOKEN VÁLIDO: Acesso a Dados Protegidos ---
Tentando acesso COM TOKEN...
Status Acesso: 200
Resposta: {'data': 'Detalhes de conta e histórico de pedidos do usuariol.', 'msg': 'Acesso concedido para: usuariol'}

=====
--- TESTE 3: TOKEN EXPIRADO (Espera de 11s) ---
=====
1. Aguardando 11 segundos para o token expirar...

--- Teste 3: Acesso Pós-Expiração: Acesso a Dados Protegidos ---
Tentando acesso COM TOKEN...
Status Acesso: 401
Resposta: {'msg': 'Token expirado. Por favor, faça login novamente.'}

=====
--- TESTE 4: TOKEN INVÁLIDO OU AUSENTE ---
=====

--- Teste 4.1: Acesso Sem Token: Acesso a Dados Protegidos ---
Tentando acesso SEM TOKEN...
Status Acesso: 401
Resposta: {'msg': 'Token não fornecido'}

--- Teste 4.2: Acesso com Token Adulterado (Assinatura Inválida) ---

--- Teste 4.2: Token Adulterado: Acesso a Dados Protegidos ---
Tentando acesso COM TOKEN...
Status Acesso: 401
Resposta: {'msg': 'Assinatura do token inválida. Token adulterado ou chave incorreta.'}

=====
--- FIM DOS TESTES AUTOMATIZADOS ---
=====
(venv) allan-samuel@allan-samuel-VirtualBox:~/python$
```

Falta de segurança

Mesmo que nossos esforços tenham sido feitos como objetivo de garantir a segurança, todo o processo se mostra em um limbo de insegurança, ou seja, os dados estão muito desprotegidos, podendo, inclusive, ser acessados com facilidade por algum hacker minimamente experiente.

PRINT 1: O print abaixo mostra que o token está desprotegido, pois ao colocar ele em um decodificador de JWT, ele nos informa o usuário e algumas outras informações que não poderiam ser acessadas tão facilmente.

Get up-to-speed with JSON Web Tokens. [Get the JWT Handbook for free](#)

JWT Debugger

[Debugger](#)[Introduction](#)[Libraries](#)[Ask](#)

ENCODED VALUE☐ Enable auto-focus

JSON WEB TOKEN (JWT)

COPYCLEAR

Valid JWT

Invalid Signature

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJleHAiOjE3NjI0MzAwMjYsImhhdCI6MTkzMDM0MDAxNiwiOiJ0dXN1YXJpbzEiLCJyb2xiOiJ0dXN1YXJ9.zYQA9WFwb0JXKFvpn8VPeMvB9a7-4LE52q_sja0TLik

DECODED HEADER

JSONCLAIMS TABLE

COPY

```
{  "alg": "HS256",  "typ": "JWT"}
```

DECODED PAYLOAD

JSONCLAIMS TABLE

COPY

```
{  "exp": 1762430026,  "iat": 1762430016,  "sub": "usuario1",  "role": "user"}
```

JWT SIGNATURE VERIFICATION (OPTIONAL)

Enter the secret used to sign the JWT below:

SECRET

COPYCLEAR

signature verification failed

a-string-secret-at-least-256-bits-long

PRINT 2: O print abaixo é ainda mais grave, ao verificar o pacote de POST do login, observamos que o usuário e a senha estão totalmente sem proteção, podendo uma simples snifada de pacotes observar o que foi dito.

JavaScript Object Notation: application/json															
0000	00 00 00 00 00 00 00 00	00 00 00 00 08 00 45 00	E.												
0010	00 64 5c 6e 40 00 40 06	e0 23 7f 00 00 01 7f 00	.d\n@.@.#.....												
0020	00 01 e9 be 13 88 c7 94	bb ae 10 2d 1d 25 80 18	-%..												
0030	02 00 fe 58 00 00 01 01	08 0a c9 f0 79 9f c9 f0	...X....y...												
0040	79 9f 7b 22 75 73 65 72	6e 61 6d 65 22 3a 20 22	y:{"user name": "												
0050	75 73 75 61 72 69 6f 31	22 2c 20 22 70 61 73 73	usuario1 ", "pass												
0060	77 6f 72 64 22 3a 20 22	73 65 6e 68 61 31 32 33	word": " senha123												
0070	22 7d		"}												

PRINTS DO WIRESHARK: Os diversos prints abaixo demonstram tudo o que foi dito anteriormente, mas através da perspectiva da captura de pacotes, desde o acesso aos dados restritos até a circunstância peculiar.

PRINT 1: Tentativa de acessar dados protegidos – 200 OK

/tmp/wireshark_loTTX9E3.pcapng 466 total packets, 212 shown

No.	Time	Source	Destination	Protocol	Length	Info
55	69.482837044	127.0.0.1	127.0.0.1	TCP	66	5001 → 58466 [ACK]
Seq=1 Ack=348 Win=65152 Len=0 TSval=3387980196 TSecr=3387980196						
Frame 55: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5001, Dst Port: 58466, Seq: 1, Ack: 348, Len: 0						
56	69.485012909	127.0.0.1	127.0.0.1	TCP	232	5001 → 58466 [PSH, ACK]
Seq=1 Ack=348 Win=65536 Len=166 TSval=3387980198 TSecr=3387980196 [TCP segment of a reassembled PDU]						
Frame 56: 232 bytes on wire (1856 bits), 232 bytes captured (1856 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5001, Dst Port: 58466, Seq: 1, Ack: 348, Len: 166						
57	69.485023815	127.0.0.1	127.0.0.1	TCP	66	58466 → 5001 [ACK]
Seq=348 Ack=167 Win=65408 Len=0 TSval=3387980198 TSecr=3387980198						
Frame 57: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 58466, Dst Port: 5001, Seq: 348, Ack: 167, Len: 0						
58	69.485040605	127.0.0.1	127.0.0.1	HTTP/JSON	185	HTTP/1.1 200 OK , JSON
(application/json)						
Frame 58: 185 bytes on wire (1480 bits), 185 bytes captured (1480 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5001, Dst Port: 58466, Seq: 167, Ack: 348, Len: 119						
[2 Reassembled TCP Segments (285 bytes): #56(166), #58(119)]						
Hypertext Transfer Protocol						
HTTP/1.1 200 OK\r\n						
Server: Werkzeug/3.1.3 Python/3.12.3\r\n						
Date: Thu, 06 Nov 2025 11:28:21 GMT\r\n						
Content-Type: application/json\r\n						
Content-Length: 119\r\n						
Connection: close\r\n						
\r\n						
[HTTP response 1/1]						
[Time since request: 0.002208593 seconds]						
[Request in frame: 54]						
[Request URI: http://127.0.0.1:5001/protected]						
File Data: 119 bytes						
JavaScript Object Notation: application/json						
59	69.485045527	127.0.0.1	127.0.0.1	TCP	66	58466 → 5001 [ACK]
Seq=348 Ack=286 Win=65408 Len=0 TSval=3387980198 TSecr=3387980198						
Frame 59: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 58466, Dst Port: 5001, Seq: 348, Ack: 286, Len: 0						
60	69.485459142	127.0.0.1	127.0.0.1	TCP	66	58466 → 5001 [FIN, ACK]
Seq=348 Ack=286 Win=65536 Len=0 TSval=3387980199 TSecr=3387980198						
Frame 60: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 58466, Dst Port: 5001, Seq: 348, Ack: 286, Len: 0						
61	69.485846467	127.0.0.1	127.0.0.1	TCP	66	5001 → 58466 [FIN, ACK]
Seq=286 Ack=349 Win=65536 Len=0 TSval=3387980199 TSecr=3387980199						
Frame 61: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5001, Dst Port: 58466, Seq: 286, Ack: 349, Len: 0						
62	69.485862726	127.0.0.1	127.0.0.1	TCP	66	58466 → 5001 [ACK]
Seq=349 Ack=287 Win=65536 Len=0 TSval=3387980199 TSecr=3387980199						
Frame 62: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 58466, Dst Port: 5001, Seq: 349, Ack: 287, Len: 0						

/tmp/wireshark_loTTX9E3.pcapng 466 total packets, 212 shown

No.	Time	Source	Destination	Protocol	Length	Info
63	69.486484403	127.0.0.1	127.0.0.1	TCP	74	58478 → 5001 [SYN]
Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=3387980200 TSecr=0 WS=128						
Frame 63: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 58478, Dst Port: 5001, Seq: 0, Len: 0						
No.	Time	Source	Destination	Protocol	Length	Info
64	69.486496953	127.0.0.1	127.0.0.1	TCP	74	5001 → 58478 [SYN, ACK]
Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=3387980200 TSecr=3387980200 WS=128						
Frame 64: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5001, Dst Port: 58478, Seq: 0, Ack: 1, Len: 0						
No.	Time	Source	Destination	Protocol	Length	Info
65	69.486504932	127.0.0.1	127.0.0.1	TCP	66	58478 → 5001 [ACK]
Seq=1 Ack=1 Win=65536 Len=0 TSval=3387980200 TSecr=3387980200						
Frame 65: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 58478, Dst Port: 5001, Seq: 1, Ack: 1, Len: 0						
No.	Time	Source	Destination	Protocol	Length	Info
66	69.486688994	127.0.0.1	127.0.0.1	HTTP	220	GET /protected HTTP/1.1
Frame 66: 220 bytes on wire (1760 bits), 220 bytes captured (1760 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 58478, Dst Port: 5001, Seq: 1, Ack: 1, Len: 154						
Hypertext Transfer Protocol						
GET /protected HTTP/1.1\r\n						
Host: 127.0.0.1:5001\r\n						
User-Agent: python-requests/2.32.5\r\n						
Accept-Encoding: gzip, deflate\r\n						
Accept: */*\r\n						
Connection: keep-alive\r\n						
\r\n						
[Full request URI: http://127.0.0.1:5001/protected]						
[HTTP request 1/1]						
[Response in frame: 70]						

PRINT 2: Acessar dados protegidos – 200 OK

/tmp/wireshark_loTTX9E3.pcapng 466 total packets, 212 shown

```
No.      Time      Source      Destination    Protocol Length Info
 51 69.482787094 127.0.0.1    127.0.0.1      TCP           74      58466 → 5001 [SYN]
Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=3387980196 TSecr=0 WS=128
Frame 51: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 58466, Dst Port: 5001, Seq: 0, Len: 0
No.      Time      Source      Destination    Protocol Length Info
 52 69.482797299 127.0.0.1    127.0.0.1      TCP           74      5001 → 58466 [SYN, ACK]
Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=3387980196 TSecr=3387980196 WS=128
Frame 52: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 5001, Dst Port: 58466, Seq: 0, Ack: 1, Len: 0
No.      Time      Source      Destination    Protocol Length Info
 53 69.482805829 127.0.0.1    127.0.0.1      TCP           66      58466 → 5001 [ACK]
Seq=1 Ack=1 Win=65536 Len=0 TSval=3387980196 TSecr=3387980196
Frame 53: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 58466, Dst Port: 5001, Seq: 1, Ack: 1, Len: 0
No.      Time      Source      Destination    Protocol Length Info
 54 69.482832012 127.0.0.1    127.0.0.1      HTTP         413      GET /protected HTTP/1.1
Frame 54: 413 bytes on wire (3304 bits), 413 bytes captured (3304 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 58466, Dst Port: 5001, Seq: 1, Ack: 1, Len: 347
Hypertext Transfer Protocol
GET /protected HTTP/1.1\r\n
Host: 127.0.0.1:5001\r\n
User-Agent: python-requests/2.32.5\r\n
Accept-Encoding: gzip, deflate\r\n
Accept: */*\r\n
Connection: keep-alive\r\n
Authorization: Bearer
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJleHAiOjE3NjI0Mjg4MDEsImhhdCI6MTc2MjQyODUwMScwV3ViIjoidXN1YXJpbzEiLC
\r\n
[Full request URI: http://127.0.0.1:5001/protected]
[HTTP request 1/1]
[Response in frame: 58]
```

PRINT 3: Acessar dados protegidos – 401 - UNAUTHORIZED

/tmp/wireshark_loTTX9E3.pcapng 466 total packets, 212 shown

No.	Time	Source	Destination	Protocol	Length	Info
67	69.486695309	127.0.0.1	127.0.0.1	TCP	66	5001 → 58478 [ACK]
Seq=1 Ack=155 Win=65408 Len=0 TSval=3387980200 TSecr=3387980200						
Frame 67: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5001, Dst Port: 58478, Seq: 1, Ack: 155, Len: 0						
68	69.487557153	127.0.0.1	127.0.0.1	TCP	241	5001 → 58478 [PSH, ACK]
Seq=1 Ack=155 Win=65536 Len=175 TSval=3387980201 TSecr=3387980200 [TCP segment of a reassembled PDU]						
Frame 68: 241 bytes on wire (1928 bits), 241 bytes captured (1928 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5001, Dst Port: 58478, Seq: 1, Ack: 155, Len: 175						
69	69.487570705	127.0.0.1	127.0.0.1	TCP	66	58478 → 5001 [ACK]
Seq=155 Ack=176 Win=65408 Len=0 TSval=3387980201 TSecr=3387980201						
Frame 69: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 58478, Dst Port: 5001, Seq: 155, Ack: 176, Len: 0						
70	69.487584919	127.0.0.1	127.0.0.1	HTTP/JSON	106	HTTP/1.1 401
UNAUTHORIZED, JSON (application/json)						
Frame 70: 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5001, Dst Port: 58478, Seq: 176, Ack: 155, Len: 40						
[2 Reassembled TCP Segments (215 bytes): #68(175), #70(40)]						
Hypertext Transfer Protocol						
HTTP/1.1 401 UNAUTHORIZED\r\n						
Server: Werkzeug/3.1.3 Python/3.12.3\r\n						
Date: Thu, 06 Nov 2025 11:28:21 GMT\r\n						
Content-Type: application/json\r\n						
Content-Length: 40\r\n						
Connection: close\r\n						
\r\n						
[HTTP response 1/1]						
[Time since request: 0.000895925 seconds]						
[Request in frame: 66]						
[Request URI: http://127.0.0.1:5001/protected]						
File Data: 40 bytes						
JavaScript Object Notation: application/json						
71	69.487589791	127.0.0.1	127.0.0.1	TCP	66	58478 → 5001 [ACK]
Seq=155 Ack=216 Win=65408 Len=0 TSval=3387980201 TSecr=3387980201						
Frame 71: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 58478, Dst Port: 5001, Seq: 155, Ack: 216, Len: 0						
72	69.487900527	127.0.0.1	127.0.0.1	TCP	66	58478 → 5001 [FIN, ACK]
Seq=155 Ack=216 Win=65536 Len=0 TSval=3387980201 TSecr=3387980201						
Frame 72: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 58478, Dst Port: 5001, Seq: 155, Ack: 216, Len: 0						

PRINT 4: Tentar acessar dados protegidos a partir de Token inválido e expirado – 401 UNAUTHORIZED.

/tmp/wireshark_loTTX9E3.pcapng 466 total packets, 212 shown

No.	Time	Source	Destination	Protocol	Length	Info
435	1595.154572077	127.0.0.1	127.0.0.1	HTTP	220	GET /protected HTTP/1.1
Frame 435: 220 bytes on wire (1760 bits), 220 bytes captured (1760 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 42018, Dst Port: 5001, Seq: 1, Ack: 1, Len: 154						
Hypertext Transfer Protocol						
GET /protected HTTP/1.1\r\n						
Host: 127.0.0.1:5001\r\n						
User-Agent: python-requests/2.32.5\r\n						
Accept-Encoding: gzip, deflate\r\n						
Accept: */*\r\n						
Connection: keep-alive\r\n						
\r\n						
[Full request URI: http://127.0.0.1:5001/protected]						
[HTTP request 1/1]						
[Response in frame: 439]						
436	1595.154577250	127.0.0.1	127.0.0.1	TCP	66	5001 → 42018 [ACK]
Seq=1 Ack=155 Win=65408 Len=0 TSval=3389505868 TSecr=3389505868						
Frame 436: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5001, Dst Port: 42018, Seq: 1, Ack: 155, Len: 0						
437	1595.156054094	127.0.0.1	127.0.0.1	TCP	241	5001 → 42018 [PSH, ACK]
Seq=1 Ack=155 Win=65536 Len=175 TSval=3389505869 TSecr=3389505868 [TCP segment of a reassembled PDU]						
Frame 437: 241 bytes on wire (1928 bits), 241 bytes captured (1928 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5001, Dst Port: 42018, Seq: 1, Ack: 155, Len: 175						
438	1595.156067446	127.0.0.1	127.0.0.1	TCP	66	42018 → 5001 [ACK]
Seq=155 Ack=176 Win=65408 Len=0 TSval=3389505869 TSecr=3389505869						
Frame 438: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 42018, Dst Port: 5001, Seq: 155, Ack: 176, Len: 0						
439	1595.156080908	127.0.0.1	127.0.0.1	HTTP/JSON	106	HTTP/1.1 401
UNAUTHORIZED , JSON (application/json)						
Frame 439: 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 5001, Dst Port: 42018, Seq: 176, Ack: 155, Len: 40						
[2 Reassembled TCP Segments (215 bytes): #437(175), #439(40)]						
Hypertext Transfer Protocol						
HTTP/1.1 401 UNAUTHORIZED\r\n						
Server: Werkzeug/3.1.3 Python/3.12.3\r\n						
Date: Thu, 06 Nov 2025 11:53:47 GMT\r\n						
Content-Type: application/json\r\n						
Content-Length: 40\r\n						
Connection: close\r\n						
\r\n						
[HTTP response 1/1]						
[Time since request: 0.001508831 seconds]						
[Request in frame: 435]						
[Request URI: http://127.0.0.1:5001/protected]						
File Data: 40 bytes						
JavaScript Object Notation: application/json						
440	1595.156085489	127.0.0.1	127.0.0.1	TCP	66	42018 → 5001 [ACK]
Seq=155 Ack=216 Win=65408 Len=0 TSval=3389505869 TSecr=3389505869						
Frame 440: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 42018, Dst Port: 5001, Seq: 155, Ack: 216, Len: 0						
441	1595.156445093	127.0.0.1	127.0.0.1	TCP	66	42018 → 5001 [FIN, ACK]
Seq=155 Ack=216 Win=65536 Len=0 TSval=3389505870 TSecr=3389505869						
Frame 441: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0						
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)						
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1						
Transmission Control Protocol, Src Port: 42018, Dst Port: 5001, Seq: 155, Ack: 216, Len: 0						

/tmp/wireshark_loTTX9E3.pcapng 466 total packets, 212 shown

```
442 1595.156499142 127.0.0.1 127.0.0.1 TCP 66 5001 → 42018 [FIN, ACK]
Seq=216 Ack=156 Win=65536 Len=0 TSval=3389505870 TSecr=3389505870
Frame 442: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 5001, Dst Port: 42018, Seq: 216, Ack: 156, Len: 0
No. Time Source Destination Protocol Length Info
443 1595.156511823 127.0.0.1 127.0.0.1 TCP 66 42018 → 5001 [ACK]
Seq=156 Ack=217 Win=65536 Len=0 TSval=3389505870 TSecr=3389505870
Frame 443: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 42018, Dst Port: 5001, Seq: 156, Ack: 217, Len: 0
```

Respondendo perguntas do roteiro

1. Como o JWT garante a integridade dos dados?

O servidor usa o header e o payload (que são apenas codificados em base64 e não criptografados) junto com uma chave secreta (conhecida apenas pelo servidor) para criar um hash criptográfico. Essa assinatura é anexada ao token. Quando o token retorna ao servidor o servidor refaz o hash usando a chave secreta. Se o hash criado for diferente da assinatura original, o servidor sabe que o header ou o payload foram modificados em trânsito, e o token é rejeitado.

2. Quais são as vantagens de usar JWT em vez de sessões tradicionais?

Algumas características são: Não precisar de armazenamento de sessão no servidor. A segunda é que o token é autônomo, ou seja, contém todas as informações (como vimos no laboratório). A validação é feita através da assinatura.

3. Quais medidas de segurança adicionais podem ser implementadas?

Como vimos no laboratório, o payload de POST do login é totalmente legível, ou seja, é possível ler ele com bastante facilidade, então o uso do HTTPS se mostra de extrema importância para a criptografia do token. A segunda medida de segurança é evitar colocar informações sensíveis nos payloads. A terceira medida é usar tokens de acesso de curta duração (tipo 5 minutos) para reduzir o tempo que um token fique ativo correndo risco de ser resgatado.

4. Como lidar com o refresh de tokens expirados?

Há algumas maneiras de lidar com o refresh de tokens expirados, duas delas são: A primeira delas é o Access Token, que é o JWT normal, de curta duração (tipo 15 minutos). É usado para acessar os dados protegidos (protected). A segunda maneira é utilizando o Refresh Token, que é um token separado, de longa duração (tipo 30 dias), que é armazenado de forma mais segura (como em um banco de dados). Ele não é usado para acessar dados, apenas para obter um novo token de Acesso.

Conclusão

Uma experiência interessante, apesar de complexa. O laboratório demonstrou que a segurança é um tema complexo e bastante delicado, pois deve ser feito com bastante diligência e utilizando os processos corretos. O que mais surpreende é que com conhecimentos tão básicos é possível verificar usuário e senhas de clientes desprotegidos, isso gera uma reflexão, porque pessoas maliciosas tendem a ter um conhecimento muito superior à desse laboratório, podendo praticar seus crimes cibernéticos com bastante liberdade. Isso implica em uma melhora na busca do conhecimento em segurança para que seja possível combater tudo isso.